I Have No Words And I Must Design

By Greg Costikyan

There's a lot of different kinds of games out there. A *helluva* lot. Cart-based, computer, CD-ROM, network, arcade, PBM, PBEM, mass-market adult, wargames, card games, tabletop RPGs, LARPs, freeforms. And hell, don't forget paintball, virtual reality, sports and the horses. It's all gaming.

Do these things have anything at all in common? What is a game? How can you tell a good one from a bad one?

Well, we can all do the latter: 'Good game, Joe,' you say, as you leap the net. Or put away the counters. Or reluctantly hand over your Earth Elemental card. Or divvy up the treasure. Unfortunately that's no better than saying, 'Good book,' as you turn the last page. It may be true, but it doesn't help you to write a better one. As game designers, we need a way to analyse games, to try to understand them, and to understand what works and what makes them interesting.

We need a critical language. And since this is basically a new form, despite its tremendous growth and staggering diversity, we need to invent one.

What Is a Game, Anyhow?

It's not a puzzle

In *The Art of Computer Game Design*, Chris Crawford contrasts what he call 'games' with 'puzzles'. Puzzles are static; they present the 'player' with a logic structure to be solved with the assistance of clues. 'Games', by contrast, are not static, but change with the players' actions. Some puzzles are obviously so; no one would call a crossword a 'game'. However, according to Crawford, some 'games' are really just puzzles: *Zork*, for instance. The game's sole objective is the solution of puzzles; finding objects and using them in particular ways to cause desired changes in the game-state. There is no opposition, there is no role-playing, and there are no resources to manage; victory is solely a consequence of puzzle-solving.

To be sure, *Zork* is not entirely static; the character moves from setting to setting, allowable actions vary by setting and inventory changes

with action. We must think of a continuum, rather than a dichotomy; if a crossword is 100% puzzle, *Zork* is 90% puzzle and 10% game.

Almost every game has some degree of puzzle-solving. Even a pure military strategy game requires players to solve the puzzle of making an optimum attack at *this* point with *these* units. To eliminate puzzle-solving entirely would require a game that was almost entirely exploration. *Just Grandma and Me* (a CD-ROM interactive storybook with game-like elements of decision-making and exploration) is a good example. Clicking objects on the screen causes entertaining sounds and animations, but there's nothing to 'solve': in fact, no strategy whatsoever.

A puzzle is static. A game is interactive.

It's not a toy

According to Will Wright, his *Sim City* is not a game at all, but a toy. Wright offers a ball as an illuminating comparison. It offers many interesting behaviours which you may explore. You can bounce it, twirl it, throw it, dribble it. If you wish, you may use it in a game: soccer, basketball or whatever; but the game is not intrinsic in the toy, it is a set of player-defined objectives overlaid on the toy. Just so Sim City. Like many computer games, it creates a world which the player may manipulate, but unlike a real game it provides no objective. You may choose one: to see if you can build a city without slums, perhaps. But Sim City itself has no victory conditions, no goals; it is a software toy.

A toy is interactive. A game has goals.

It's not a story

Again and again, we hear about story; interactive literature; creating a story through role-play. The idea that games have something to do with stories has such a hold on designers' imaginations that it probably can't be expunged. It deserves at least to be challenged.

Stories are inherently linear. However much characters may agonize over the decisions they make, they make them in the same way every time we reread the story, and the outcome is always the same. Indeed, this is a strength; the author chose precisely those characters, those events, those decisions and that outcome because it made for the strongest story. If the characters did something else, the story wouldn't be as interesting.

Games are inherently non-linear. They depend on decision-making. Decisions have to pose real, plausible alternatives, or they aren't real decisions. It must be entirely reasonable for a player to make a decision one way in one game, and a different way in the next. To the degree that a game is made more like a story—more linear, fewer real options—it is made less like a game.

Consider: you buy a book, or see a movie, because it has a great story. How would you react if your game referee were to tell you: 'I don't want

you players to do that, because it will ruin the story'? This may well be right, but that's beside the point. Gaming is *not* about telling stories.

That said, games often, and fruitfully, borrow elements from fiction. Role-playing games depend on characters; computer adventures and LARPs are often driven by plots. The notion of increasing narrative tension is a useful one for any game that comes to a definite conclusion. To try to hew too closely to a storyline, however, is to limit players' freedom of action and their ability to make meaningful decisions.

The hypertext fiction movement is interesting in this respect. Hypertext is inherently non-linear, so that the traditional narrative is wholly inappropriate to a hypertext work. Writers of hypertext fiction are trying to explore the nature of human existence, as does the traditional story, but in a way that permits multiple viewpoints, temporal leaps, and reader construction of the experience. Hypertext writers share something—more than they know— with games designers, and also something with the writers of traditional narrative; but if hypertext fiction ever becomes artistically successful (nothing I've read so far is), it will be through the creation of a new narrative form, something that we will be hard-pressed to call a 'story'.

Stories are linear. Games are not.

It demands participation

In a traditional art-form, the audience is passive. When you look at a painting you may imagine things in it, you may see something other than what the artist intended, but your role in constructing the experience is slight. The artist painted. You see. You are passive.

When you go to the movies, or watch TV, or visit the theatre, you sit and watch and listen. Again, you do interpret, to a degree; but you are the audience. You are passive. The art is created by others.

When you read a book, most of it goes on in your head and not on the page; but still—you're receiving the author's words. You're passive.

It's all too, too autocratic: the mighty artist condescends to share their genius with lesser mortals. How can it be that, two hundred years after the American War of Independence, we still have such aristocratic forms? Surely we need forms in spirit with the times; forms which permit the common man to create his own artistic experience.

Enter the game. Games provide a set of rules; but the players use them to create their own consequences. It's something like the music of John Cage: he wrote themes around which the musicians were expected to improvise. Games are like that; the designer provides the theme, the players the music. A democratic art-form for a democratic age.

Traditional artforms play to a passive audience. Games require active participation.

What Is a Game?

A game is a form of art in which participants, termed players, make *decisions* in order to manage *resources* through *game tokens* in the pursuit of a *goal*.

Decision-making

I offer this term in an effort to destroy the inane and overhyped word 'interactive'. The future, we are told, will be interactive. You might as well say, 'The future will be fnurglewitz.' It would be about as enlightening. A light switch is interactive. You flick it up, the light turns on. You flick it down, the light turns off. That's interaction, but it's not a lot of fun.

All games are interactive. The game-state changes with the players' actions. If it didn't, it wouldn't be a game; it would be a puzzle. But 'interaction' has no value in itself. Interaction must have purpose.

Suppose we have a product that's interactive. At some point, you are faced with a choice: You may choose to do A, or to do B. What makes A better than B? Or is B better than A at some times but not at others? What factors go into the decision? What resources are to be managed? What's the eventual goal?

Aha! Now we're not talking about 'interaction'. Now we're talking about decision-making.

The thing that makes a game a game is the need to make decisions. Consider *Chess*[1]. It has few of the aspects that make games appealing: no element of simulation, no role-playing and damn little colour. What it's got is the need to make decisions. The rules are tightly constrained, the objectives clear, and victory requires you to think several moves ahead. Excellence in decision-making is what brings success.

What do players do in any game? Some things depend on the medium. In some games, they roll dice. In some games, they chats with their friends. In some games, they whack at a keyboard. But in every game, they make decisions.

At every point, they consider the game-state. That might be what they see on the screen. Or it might be what the referee has just told them. Or it might be the arrangement on the pieces on the board. Then they consider their objectives, and the game tokens and resources available to them. They consider their opposition, the forces they must struggle against. They try to decide on the best course of action.

And they make a decision.

What's the key here? Goals. Opposition. Resource management. Information. We'll talk about them in a moment.

What decisions do players make in this game?

Goals

Sim City has no goals. Is it not a game?

No, as its own designer willingly maintains. It is a toy.

The only way to stay interested in it for very long is to turn it into a game by setting goals, by defining objectives for yourself. Build the grandest possible megalopolis; maximize how much your people love you; build a city that relies solely on mass transit. Whatever goal you've chosen, you've turned it into a game.

Even so, the software doesn't support your goal. It wasn't designed with your goal in mind. Trying to do something with a piece of software that it wasn't intended for can be awfully frustrating.

Since there's no goal, *Sim City* soon palls. By contrast, Sid Meier and Bruce Shelley's *Civilization*—an obviously derivative product—has explicit goals and is far more involving and addictive.

'But what about role-playing games?' you may say. 'They have no victory conditions.'

No victory conditions, true, but certainly they have goals; lots of them. Rack up the old experience points. Fulfil the quest your friendly referee has just inflicted on you. Rebuild the Imperium and stave off civilization's final collapse. Strive toward spiritual perfection. Whatever.

If, for some reason, your player characters don't have a goal, they'll find one right quick. Otherwise, they'll have nothing better to do than sit around the tavern and grouse about how boring the game is. Until the referee gets pissed off and has a bunch of orcs show up and try to beat their heads in. Now they've got a goal. Personal survival is a good goal. One of the best.

If you have no goal, decisions are meaningless. Choice A is as good as Choice B; pick a card, any card. Who cares? What does it matter?

For it to matter, for the game to be meaningful, you need something to strive toward. You need goals.

What are the players' goals? Can the game support a variety of different goals? What facilities exist to allow players to strive toward their various goals?

Opposition

Oh, say the politically correct. Those bad, icky games. They're so competitive. Why can't we have co-operative games?

'Co-operative games' generally seem to be variants of 'let's all throw a ball around'. Oh golly, how fascinating, I'll stop playing *Mortal Kombat* for that, you betcha.

Are we really talking about *competition*?

Yes and no; many players do get a kick out of beating others with their naked minds alone, which is at least better than naked fists. Chess players are particularly obnoxious in this regard. Still, the real interest is in struggling toward a goal.

The most important word in that sentence is: struggling.

Here's a game. It's called *Plucky Little England*, and it simulates the situation faced by the United Kingdom after the fall of France in World War II. Your goal: preserve liberty and democracy and defeat the forces of darkness and oppression. You have a choice:

A. Surrender.

B. Spit in Hitler's eye! Rule Britannia! Britons never, never, never shall be slaves!

You chose B? Congratulations! You won!

Now, wasn't that satisfying? Ah, the thrill of victory.

There is no thrill of victory, of course; it was all too easy, wasn't it? There wasn't any struggle.

In a two-player, head-to-head game, your opponent is the opposition, your struggle against him; the game is a direct competition.

And this is a first-rate way of providing opposition. Nothing is as sneaky and as hard to overcome as a determined human opponent. Direct competition isn't the only way to do it, though.

Think of fiction. The ur-story, the Standard Model Narrative, works like this: character A has a goal. He faces obstacles B, C, D, and E. He struggles with each, in turn, growing as a person as he does so. Ultimately, he overcomes the last and greatest obstacle.

Do these obstacles all need to be The Villain, The Bad Guy, The Opponent, The Foe? No, though a good villain makes for a first-rate obstacle. The forces of nature, cantankerous mothers-in-law, crashing hard drives and the hero's own feelings of inadequacy can make for good obstacles too.

Just so in games.

In most RPGs, the 'opposition' consists of non-player characters, and players are expected to co-operate with one another. In many computer games, the 'opposition' consists of puzzles you must solve. In LARPs, the 'opposition' is often the sheer difficulty of finding the player who has the clue or the widget or the special power you need. In most solitaire games, your 'opposition' is really a random element, or a set of semi-random algorithms against which you are pitted.

Whatever goals you set your players, you must make the players work to achieve their goals. Setting them against each other is one way to do that, but not the only one. Even when a player has an opponent, putting other obstacles in the game can increase its richness and emotional appeal.

The desire for 'co-operative games' is the desire for an end to strife. There can be none. Life is the struggle for survival and growth. There is no end to strife, not this side of the grave. A game without struggle is a game that's dead.

What provides opposition? What makes the game a struggle?

Managing resources

Trivial decisions aren't any fun. Remember Plucky Little England? There wasn't any real decision, was there?

Or consider Robert Harris's *Talisman*. Each turn, you roll the die. The result is the number of spaces you can move. You may move to the left, or to the right, around the track.

Well, this is a little better than a traditional track game; the player has got a choice. But 99 times out of 100, either there's no difference between the two spaces, or one is obviously better than the other. The choice is bogus.

The way to make choices meaningful is to give players resources to manage.

'Resources' can be anything. Panzer divisions. Supply points. Cards. Experience points. Knowledge of spells. Ownership of fiefs. The love of a good woman. Favours from the boss. The good will of an NPC. Money. Food. Sex. Fame. Information.

If the game has more than one 'resource', decisions suddenly become more complex. If I do this, I get money and experience, but will Lisa still love me? If I steal the food, I get to eat, but I might get caught and have my hand cut off. If I declare against the Valois, Edward Plantagenet will grant me the Duchy of Gascony, but the Pope may excommunicate me, imperilling my immortal soul.

These are not just complex decisions; these are interesting ones. Interesting decisions make for interesting games.

The resources in question have to have a game role; if 'your immortal soul' has no meaning, neither does excommunication. (Unless it reduces the loyalty of your peasants, or makes it difficult to recruit armies, or … but these are game roles, *n'est-ce pas*?) Ultimately, 'managing resources' means managing game elements in pursuit of a goal. A 'resource' that has no game role has nothing to contribute to success or failure and is ultimately void.

What resources does the player manage? Is there enough diversity in them to require trade-offs in making decisions? Do they make those decisions interesting?

Game tokens

You affect actions in the game through your game tokens. A game token is any entity you may manipulate directly. In a boardgame, it is your pieces. In a cardgame, it is your cards. In a role-playing game, it is your character. In a sports game, it is you yourself.

What is the difference between 'resources' and 'tokens'? Resources are things that must be managed efficiently to achieve the goals; tokens are the means of managing them. In a board wargame, combat strength is a resource; counters are tokens. In a role-playing game, money is a resource used through player characters.

Why is this important? Because without game tokens, you wind up with a system that operates without much player input. Will Wright and Fred Haslam's *Sim Earth* is a good example. In *Sim Earth,* the player sets some parameters and sit back to watch the game play out itself. There's very little to do, no tokens to manipulate, no resources to manage. Just a few parameters to twiddle with. This is mildly interesting, but not very.

To give players a sense that they control their destiny, that they are playing a game, you need game tokens. The fewer the tokens, the more detailed they must be. It is no coincidence that role-playing games, which give the player a single token, also have exceptionally detailed rules for what that token can do.

What are the players' tokens? What are these tokens' abilities? What resources do they use? What makes them interesting?

Information

I've had more than one conversation with a computer game designer in which he tells me about all the fascinating things his game simulates while I sit there saying, 'Really? What do you know, I didn't realize that.'

Take a computer wargame in which weather affects movement and defence. If you don't tell players that weather has an effect, what good is it? It won't affect the players' behaviour; it won't affect their decisions.

Or maybe you do tell them weather has an effect, but the players have no way of telling whether it's raining or snowing or whatever at any given time. Again, what good is that?

Or maybe they *can* tell, and they *do* know, but they have no idea what effect weather has. Maybe it cuts everyone's movement in half, or maybe it slows movement across fields to a crawl but does nothing to units moving along roads. This is better, but not a whole lot.

The interface must provide the players with relevant information. And they must have enough information to be able to make sensible decisions.

That isn't to say a player must know everything; hiding information can be very useful. It's quite reasonable to say, 'you don't know just how strong your units are until they enter combat,' but in that case the player must have some idea of the range of possibilities. It's reasonable to say, 'you don't know what card you'll get if you draw to an inside straight,' but only if the player has some idea what the odds are. If I might draw the Queen of Hearts and might draw Death and might draw the Battleship Potemkin, I have absolutely no basis on which to make a decision.

More than that, the interface must not provide too much information, especially in a time-dependent game. If weather, supply state, the mood of my commanders, the fatigue of the troops and what Tokyo Rose said on the radio last night can all affect the outcome of my next decision, and I have to decide some time in the next five seconds and

it would take me five minutes to find all the relevant information by pulling down menus and looking at screens, the information is still irrelevant. I may have access to it, but I can't reasonably act on it.

Or let's talk about computer adventures; they often display information failure. 'Oh, to get through the Gate of Thanatos, you need a hatpin to pick the lock. You can find the hatpin on the floor of the Library. It's about three pixels by two pixels, and you can see it, if your vision is good, between the twelfth and thirteenth floorboards, about three inches from the top of the screen. What, you missed it?'

Yeah, I missed it. In an adventure, it shouldn't be ridiculously difficult to find what is needed, nor should victory be impossible just because you made a wrong decision three hours and thirty-eight decision points ago. Nor should the solutions to puzzles be arbitrary or absurd.

Or consider freeforms. In a freeform, players are often given goals, and achieving them requires them to find out several things—call them Facts A, B, and C. The freeform's designer had better make damn sure that A, B, and C are out there somewhere—known to other characters, or on a card that's circulating in the game—whatever they are, they have to be there and the players have to have some chance of finding them. Otherwise the players have no chance of achieving their goals, and that's no fun.

Given the decisions players are required to make, what information do they need? Does the game provide the information as and when needed? Will reasonable players be able to figure out what information they need, and how to find it?

Other Ways To Strengthen Games

Diplomacy

Achieving a goal is meaningless if it comes without work, if there is no opposition; but that doesn't mean all decisions must be zero-sum. Whenever multiple players are involved, games are strengthened if they permit and encourage diplomacy.

Games permit diplomacy if players can assist each other—perhaps directly, perhaps by combining against a mutual foe. Not all multi-player games do this; in *Monopoly*, for instance, there's no effective way either to help or hinder anyone else. There's no point in saying, 'Let's all get Joe', or, 'Here, you're a novice, I'll help you out, you can scratch my back later,' because there's no way to do it.

Some games permit diplomacy, but not much. In Lawrence Harris's *Axis & Allies*, players can help each other to a limited degree, but everyone is permanently Axis or permanently Allied, so diplomacy is never a key element of the game.

One way to encourage diplomacy is by providing non-exclusive goals. If you're looking for the Ark of the Covenant, and I want to kill Nazis, and the Nazis have got the Ark, we can work something out. Maybe our

alliance will end when the French Resistance gets the Ark and we wind up on opposite sides, but such twists are what make games fun.

Games can encourage diplomacy even when players are directly opposed. The diplomatic game *par excellence* is, of course, Calhammer's *Diplomacy*, in which victory more often goes to the best diplomat than to the best strategist. The key to the game is the Support order, which allows one player's armies to assist another in an attack, encouraging alliance.

Alliances never last, to be sure; Russia and Austria may ally to wipe out Turkey, but only one of them can win. Eventually, one will stab the other in the back.

Fine. It's the need to find allies, retain them, and persuade enemies to change their stripes that makes sure players will keep on talking. If alliances are set in stone, diplomacy comes to an end.

Computer games are almost inherently solitaire and even when they permit diplomacy with NPC computer opponents, they generally don't make it interesting. Network games are, or ought to be, inherently diplomatic; as network games become more prevalent, we can expect most developers from the computer design community to miss this point entirely. As an example, when the planners of interactive TV networks talk about games, they almost exclusively talk about the possibility of downloading cart-based (Nintendo, Sega) games over cable. They're doing so for a business reason: billions are spent annually on cart-based games, and they'd like a piece of the action. They don't seem to realize that networks permit a wholly different kind of gaming, which has the potential to make billions in its own right—and that this is the real business opportunity.

How can players help or hinder each other? What incentives do they have to do so? What resources can they trade?

Colour

Monopoly is a game about property development. Right?

Well, no, obviously not. A property developer would laugh at the notion. A game about property development needs rules for construction loans and property syndication and union work rules and the bribery of municipal inspectors.

Monopoly has nothing to do with property development. You could take the same rules, change the board, pieces and cards, and make it into a game about space exploration, say. Except that your game would have as much to do with space exploration as *Monopoly* has to do with property development.

Monopoly isn't really about anything. But it has the 'colour' of a property game: named properties, little plastic houses and hotels, play money. And that's a big part of its appeal.

Colour counts for a lot. As a simulation of World War II, Lawrence Harris's *Axis & Allies* is a pathetic effort. Ah, but the colour! Millions of little plastic airplanes and battleships and tanks! Thundering dice! The world at war! The game works almost solely because of its colour.

Or consider Frank Chadwick's *Space: 1889*. The rules do nothing to evoke the Burroughsian wonders, the pulp-action thrills, the Kipling-esque Victorian charms to be gained from the game's setting. Despite a clean system and a detailed world, it is curiously colourless and suffers for it. Pageantry, detail and sense of place can greatly add to a game's emotional appeal.

This has almost nothing to do with the game qua game. The original Nova edition of *Axis & Allies* was virtually identical to the Milton Bradley edition. Except that it had a godawful garish paper map, some of the ugliest counters I've ever seen, and a truly amateurish box. I looked at it once, put it away and never looked at it again.

Yet the Milton Bradley edition, with all the little plastic pieces, still gets pulled out now and again. Same game. Far better colour.

How does the game evoke the ethos, atmosphere and pageantry of its setting? What can be done to make it more colourful?

Simulation

Many games simulate nothing. The oriental folk-game *Go*, say; little stones on a grid. It's abstract to perfection. Or John Horton Conway's *Life*; despite the evocative name, it's merely an exploration of a mathematical space.

Nothing wrong with that. But colour adds to a game's appeal. And simulation is a way of providing colour.

Suppose I think, for some reason, that a game about the battle of Waterloo would have great commercial appeal. I could, if I wanted, take *Monopoly*, change 'Park Lane' to 'Quatre Bras' and the hotels to plastic soldiers, and call it *Waterloo*. It would work, but wouldn't it be better to simulate the battle? To have little battalions manoeuvring over the field? To hear the thunder of guns?

Or take *Star Wars: The Role-Playing Game*, which I designed. I could have taken Gygax and Arneson's *Dungeons & Dragons* and changed it around, calling swords 'blasters' and magic-users 'Jedi'. Instead, I set out to simulate the movies, to encourage the players to attempt far-fetched cinematic stunts, to use the system itself to reflect something about the atmosphere and ethos of the films.

Simulation has further value. For one, it improves character identification. A Waterloo based on *Monopoly* would do nothing to make players think like Wellington and Napoleon; Kevin Zucker's *Napoleon's Last Battles* does much better, forcing players to think about the strategic problems those men faced.

It can allow insight into a situation that mere narrative cannot. It allows players to explore different outcomes—in the fashion of a software toy—and thereby come to a gut understanding of the simulation's subject. Having played at least a dozen different games about Waterloo, I understand the battle, why things happened the way they did, and the nature of Napoleonic warfare, far better than if I had merely read a dozen books on the subject.

Simulating something is almost always more complicated than simply exploiting a theme for colour. And it is not, therefore, for every game. But when the technique is used, it can be powerful.

How can elements of simulation strengthen the game?

Variety of encounter

'You just got lucky.'

Words of contempt; you won through the vagaries of chance. A game that permits this is obviously inferior to ones where victory goes to the skilled, smart and strong. Right?

Not necessarily.

'Random elements' in a game are never wholly random. They are random within a range of possibilities. When, in a board wargame, I make an attack, I can look at the Combat Results Table. I know what outcomes are possible, and my chances of achieving what I want to achieve. I take a calculated risk. Over the whole game, I make dozens or hundreds of die-rolls. Given so much reliance on randomness, the 'random element' regresses to a mean. Except in rare cases, my victory or defeat will be based on my excellence as a strategist, not on my luck with the dice.

Randomness can be useful. It's one way of providing variety of encounter.

What does that mean?

It means that the same old thing all over again is fucking boring. It means that players like to encounter the unexpected. It means that the game has to allow lots of different things to happen, so there's always something a little different for the players to encounter.

In a game like *Chess*, that 'something different' is the ever-changing implications of the positions of the pieces. In a game like Richard Garfield's *Magic: The Gathering*, it's the sheer variety of cards, the random order in which they appear and the interesting ways in which they can be combined. In Arneson and Gygax's *Dungeons & Dragons*, it's the staggering variety of monsters and spells, coupled with the dungeon master's ingenuity in throwing new situations at the players.

If a game has inadequate variety, it rapidly palls. That's why no one plays graphic adventures more than once; there's enough variety for a single game, but it's the same thing all over again the next time you play. That's why *Patience*, the solitaire cardgame, becomes dull pretty

fast; you're doing the same things over and over, and reshuffling the cards isn't enough to rekindle your interest, after a time.

What things do the players encounter in this game? Is there enough for them to explore and discover? What provides variety? How can the variety of encounter be increased?

Position identification

'Character identification' is a common theme of fiction. Writers want readers to like their protagonists, to identify with them, to care what happens to them. Character identification lends emotional power to a story.

The same is true in games. To the degree that players are encouraged to care about 'the side', to identify with their position in the game, the game's emotional impact is increased.

The extreme case is sports; in sports, your 'position' is you. You're out there on the baseball diamond; winning or losing matters, and you feel it deeply when you strike out, or smash the ball out of the park. It's important to you. So important that fistfights and bitter words are not uncommon in every sport. So important that we've invented a whole cultural tradition of 'sportsmanship' to try to prevent these unpleasant feelings from coming to the fore.

Role-playing games are one step abstracted; your character isn't you, but you invest a lot of time and energy in it. It's your sole token and the sum total of your position in the game. Bitter words, and even fistfights, are not unknown among role-players, though rather rarer than in sports.

Getting players to identify with their game position is straightforward when a player has a single token; it's harder when they control many. Few people feel much sadness at the loss of a knight in Chess or an infantry division in a wargame. But even here, a game's emotional power is improved if the player can be made to feel identification with 'the side'.

One way to achieve this is to make clear the player's point of view. Confusion about point of view is a common failing of boardgame design. For instance, Richard Berg's *Campaigns for North Africa* claims to be an extraordinarily realistic simulation of the Axis campaign in Africa. Yet the players spend a great deal of time worrying about the locations of individual pilots and how much water is available to individual batallions. Rommel's staff might worry about such things, but Rommel assuredly did not. Who is the player supposed to be? The accuracy of the simulation is, in a sense, undermined, not supported, by the level of detail.

What can you do to make the players care about their position? Is there a single game token that's more important than others to the players, and what can be done to strengthen identification with it? If not, what is the overall emotional

appeal of the postion, and what can be done to strengthen that appeal? Who 'are' the players in the game? What is their point of view?

Role-playing

HeroQuest has been termed a 'role-playing boardgame'. As in a role-playing game, each player controls a single character which, in *HeroQuest*'s case, is a single plastic figure on the board. If you are a single character, are you not 'playing a role'? Is the characterization of this game as a 'role-playing' game therefore justified?

No, to both questions.

The questions belie confusion between 'position identification' and 'role-playing'. I may identify closely with a game token without feeling that I am playing a role.

Role-playing occurs when, in some sense, players take on the persona of their positions. Different players and different games may do this in different ways. Perhaps they try to speak in the language and rhythm of their characters. Perhaps they talk as if they are feeling the emotions their characters feel. Perhaps they talk as they normally do, but they give serious consideration to 'what my *character* would do in this case' as opposed to 'what *I* want to do next?'

Role-playing is naturally most common in role-playing games,but it can occur in other environments, as well. I, for one, can't get through a game of Vincent Tsao's *Junta* without talking in a phoney Spanish accent somewhere along the line. The game makes me think sufficiently like a big man in a corrupt banana republic that I start to play the role.

Role-playing is a powerful technique for a whole slew of reasons. It improves position identification: if players think like their characters, they're identifying with them closely. It improves the game's colour, because the players become partly responsible for maintaining the willing suspense of disbelief, the feeling that the game world is alive and colourful and consistent. Plus it is an excellent method of socialization.

Indeed, the connection with socialization is key: role-playing is a form of performance. In a role-playing game, role-players perform for the amusement of their friends. If there aren't any friends, there's no point to it.

Which is why so-called 'computer role-playing games' are nothing of the kind. They have no more connection with role-playing than does *HeroQuest*. That is, they have the trappings of role-playing: characters, equipment and stories; but there is no mechanism for players to ham it up, to characterize themselves by their actions, to role-play in any meaningful sense.

This is intrinsic in the technology. Computer games are solitaire and solitaire gamers have, by definition, no audience. Therefore, computer games cannot involve role-playing. Add a network, and you can have a role-playing game. Hence the popularity of MUDs.

How can players be induced to role-play? What sorts of roles does the system permit or encourage?

Socializing

Historically, games have mainly been used as a way to socialize. For players of *Bridge*, *Poker*, and *Charades*, the game is secondary to the socialization that goes on over the table. One oddity of the present is that the most commercially successful games are all solitary in nature: cartridge games, disk-based computer games, CD-ROM games. Once upon a time, our image of gamers was people sitting around a table and playing cards; now, it's a solitary adolescent, twitching a joystick before a flickering screen.

Yet at the same time we see the development of role-playing, in both tabletop and live-action form, which depends utterly on socialization. And we see that the most successful mass-market boardgames, like *Trivial Pursuit* and *Pictionary*, are played almost exclusively in social settings.

I believe that the solitary nature of most computer games is a temporary aberration, a consequence of the technology, and that as networks spread and their bandwidth increases, the historical norm will reassert itself.

When designing any game, it is worthwhile thinking about the game's social uses, and how the system encourages or discourages socialization. For instance, almost every network has on-line versions of classic games like *Poker* and *Bridge*. In almost every case, those games have failed to attract much usage. The exception: America Online, which permits real-time chat between players. Their version of network *Bridge* allows for table talk, and it has been quite popular.

Or as another example, many tabletop role-playing games spend far too much effort worrying about 'realism' and far too little about the game's use by players. Of what use is a combat system that is extraordinarily realistic, if playing out a single combat round takes fifteen minutes, and a whole battle takes four hours? Players are not spending their time socializing and talking and hamming it up; they're spending time rolling dice and looking things up on charts. What's the point in that?

How can the game better encourage socialization?

Narrative tension

Nebula-award-winning author Pat Murphy says that the key element of plot is 'rising tension'. That is, a story should become more gripping as it proceeds, until its ultimate climactic resolution.

Suppose you're a Yankees fan. Of course, you want to see the Yankees win. Even so, if you go to a game at the ballpark, do you really want to see them develop a seven-point lead in the first inning and wind up winning 21–2? Yes, you want them to win, but this doesn't make for a very interesting game. What would make you rise from your seat in excite-

ment and joy is to see them pull out from behind in the last few seconds of the game with a smash home run with bases loaded. Tension makes for fun games.

Ideally, a game should be tense all the way through, but especially so at the end. The toughest problems, the greatest obstacles, should be saved for last. You can't always ensure this, especially in directly competitive games: a *Chess* game between a grandmaster and a rank beginner is not going to involve much tension. But it should be possible, especially in solitaire computer games, to ensure that every stage of the game involves a set of challenges, and that the player's job is completed only at the end.

In fact, one of the most common game failures is anticlimax. The period of maximum tension is not the resolution, but somewhere midway through the game. After a while, the opposition is on the run, or the player's position is unassailable. In most cases, this is because the designer never considered the need for narrative tension.

What can be done to make the game tense?

They're All Alike Under the Dice

We're now equipped to answer the questions I posed at the beginning of this article.

Do all the myriad forms of gaming have anything in common? Most assuredly. All involve decision-making and managing resources in pursuit of a goal; that's true whether we're talking about *Chess* or *Seventh Guest*, *Mario Brothers* or *Vampire*, Roulette or *Magic: The Gathering*. It's universal; it's what defines a game.

How can you tell a good game from a bad one? The test is still in the playing; but we now have some terms to use to analyse a game's appeal.

Chess involves complex and difficult decisions; Magic has an enormous variety of encounters; *Roulette* has an extremely compelling goal (money—the real stuff). More detailed analysis is possible, to be sure, and is left as an exercise for the reader.

Is the analytical theory presented here hermetic and complete? Assuredly not; there are games that defy many, though not all, of its conclusions (e.g. *Candyland*, which involves no decision-making whatsoever). No doubt there are aspects to the appeal of games it overlooks.

It is to be considered a work in progress: a first stab at codifying the intellectual analysis of the art of game design. Others are welcome, even encouraged, to build on its structure—or to propound alternative theories in its defiance. If we are to produce works worthy to be termed 'art', we must start to think about what it takes to do so, to set ourselves goals beyond the merely commercial. For we are embarked on a voyage of revolutionary import: the democratic transformation of the arts. Properly addressed, the voyage will lend grandeur to our civilization; improperly,

it will create merely another mediocrity of the TV age, another form wholly devoid of intellectual merit.

Greg Costikyan has designed twenty-three commercially published games, including five winners of the Origins Award. Among them are Star Wars: The Role-Playing Game, Paranoia, Toon, The Creature That Ate Sheboygan *and* Madmaze. *He is also the author of three novels and numerous short stories, and resides in Jersey City with two computers and a hamster. Comments on his article are welcome, and may be directed to costik@ritz.mordor.com.*

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Note

[1] In normal practice, the names of traditional games, e.g. chess, go, poker, are uncapitalized, as is usual with common nouns. The names of proprietary games are written with Initial Caps. This usage is inconsistent with the thesis that games are an art-form, and that each game, regardless of its origins, must be viewed as an oeuvre. I capitalize and use italics for all game names, throughout the article. We capitalize *Beowulf*, though it is the product of folk tradition rather than a definite author, just as we capitalize *One Hundred Years of Solitude*. In the same fashion, I capitalize *Chess*, though it is the product of folk tradition rather than a definite designer, just as I capitalize *Dungeons & Dragons*.